Experiment No: 8

Title : Study of packet sniffer tools wireshark: -
a. Observer performance in promiscuous as well as non-promiscuous mode.
b. Show the packets can be traced based on different filters.

Aim
To capture and analyze network packets using Wireshark, observe packet capturing in promiscuous and non-promiscuous modes, and trace network packets using different filters.

Tools / Requirements
Operating System: Microsoft Windows

Packet Analyzer Tool: Wireshark

Internet connection

Web browser for generating network traffic

Theory
A packet sniffer is a tool used to capture and analyze network packets transmitted over a network. These packets contain information such as source address, destination address, protocol type, and data payload

Wireshark is a network protocol analyzer that allows users to capture live network traffic and examine packets in detail. It helps network administrators and security professionals monitor network communication and troubleshoot network issues.

When data is transmitted over a network, it is divided into small units called packets. Each packet contains:

• Source IP address

• Destination IP address

• Protocol type (TCP, UDP, HTTP, DNS etc.)

• Packet length

• Data payload

Wireshark displays captured packets in three sections:

1. Packet List Pane – Shows all captured packets.

2. Packet Details Pane – Displays protocol details of a selected packet.

3. Packet Bytes Pane – Shows raw packet data.

Promiscuous Mode:

In promiscuous mode, the network interface captures all packets passing through the network regardless of whether they are addressed to the local machine.

Non-Promiscuous Mode:

In non-promiscuous mode, the network interface captures only packets addressed to the local computer.

Packet Filtering:

Wireshark provides display filters that allow users to view specific packets from the captured traffic. Filters help analyze particular protocols or packets from specific hosts.

Procedure
Step 1 – Install Wireshark

1. Download Wireshark from https://www.wireshark.org

2. Install the software.

3. Allow installation of Npcap driver during setup.

Part A – Observing Promiscuous Mode

Step 2 – Start Wireshark

1. Open Wireshark.

2. Select the active network interface such as Wi-Fi or Ethernet.

Step 3 – Enable Promiscuous Mode

1. Click Capture → Options.

2. Ensure Promiscuous Mode checkbox is enabled.

3. Click Start to begin capturing packets.

Step 4 – Generate Network Traffic

Open a browser and visit websites such as:

www.google.com

www.facebook.com

www.youtube.com

Step 5 – Stop Packet Capture

Click the Stop button in Wireshark.

Observation: A large number of packets from different protocols are captured because promiscuous mode captures all network traffic.

Part B – Observing Non-Promiscuous Mode

Step 6 – Disable Promiscuous Mode

1. Go to Capture → Options.

2. Uncheck Enable promiscuous mode.

3. Start packet capture again.

Step 7 – Generate Network Traffic Again

Open any website again.

Observation: Only packets related to the local system are captured and fewer packets are displayed.

Part C – Packet Tracing Using Filters

Filter 1 – DNS Packets

Dns

Filter 2 – HTTP Packets

http

Filter 3 – TCP Packets

tcp

Filter 4 – Packets from Specific IP Address

ip.addr == 8.8.8.8

Filter 5 – Website Domain Detection

tls.handshake.extensions_server_name

Observation Table

Promiscuous Mode – Captures all packets in the network

Non-Promiscuous Mode – Captures only packets addressed to the local machine

DNS Filter – Displays DNS queries and responses

HTTP Filter – Displays HTTP packets

TCP Filter – Displays TCP packets

IP Filter – Displays packets related to a specific IP address

Result

The packet sniffer tool Wireshark was successfully used to capture and analyze network packets. Packet capture was observed in both promiscuous and non-promiscuous modes, and packets were traced using different display filters.

Conclusion

Wireshark is an effective network protocol analyzer used to monitor and analyze network traffic. By using promiscuous and non-promiscuous modes along with packet filters, network packets can be captured and analyzed efficiently.